

CHILDHOOD RHEUMATIC DISEASES

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OUTLINE

- Introduction on Rheumatic diseases of childhood
(Connective Tissue Disease, Collagen Vascular Diseases)
- Juvenile Idiopathic Arthritis
- SLE

Introduction

- Are defined by the constellation of results of the **physical examination, autoimmune marker and other serologic tests, tissue pathology & imaging.**
- **Defined diagnostic criteria** exist for most rheumatic diseases.
- Recognition of **clinical patterns remains essential for diagnosis** because there is **no single diagnostic test.**
- The primary mimics of rheumatic diseases are **infection** and **malignancy** but also include metabolic, orthopedic, immune deficiencies, autoinflammatory diseases & chronic pain conditions.
- Exclusion of possible mimicking disorders is essential before initiation of treatment for a presumptive diagnosis, especially corticosteroids.

Symptoms Suggestive of Rheumatic Disease

- There are no classic symptoms of a rheumatic disease, but common symptoms include:-
 - ✓ Joint pain
 - ✓ Fever
 - ✓ Fatigue
 - ✓ Rash.
- Presenting signs and symptoms help direct the evaluation and limit unnecessary testing.

Conti...

- **Arthralgias** are common in childhood
- Although rheumatic diseases may manifest as arthralgias
 - ✓ **Arthritis is a stronger predictor** of the presence of rheumatic disease.
- **Arthralgias without** physical findings for **arthritis** suggest **infection, malignancy, orthopedic conditions.**

Conti...

- **Arthralgia** may also be a presenting symptom of pediatric **SLE** and juvenile idiopathic arthritis (**JIA**).

- Other symptoms more suggestive of **arthritis** include:-
 - ✓ **Morning stiffness**
 - ✓ **Joint swelling**
 - ✓ **limited range of motion**
 - ✓ **Pain with joint motion**
 - ✓ **Gait disturbance**
 - ✓ **Fatigue or stiffness after physical inactivity (gelling phenomenon).**

Table 194.1 **Symptoms Suggestive of Rheumatic Disease**

SYMPTOM	RHEUMATIC DISEASE(S)	POSSIBLE NONRHEUMATIC DISEASES CAUSING SIMILAR SYMPTOMS
Fevers	Systemic JIA, SLE, vasculitis, acute rheumatic fever, sarcoidosis, MCTD	Malignancies, infections and postinfectious syndromes, inflammatory bowel disease, periodic fever (autoinflammatory) syndromes, Kawasaki disease, HSP*
Arthralgias	JIA, SLE, rheumatic fever, JDM, vasculitis, scleroderma, sarcoidosis	Hypothyroidism, trauma, endocarditis, other infections, pain syndromes, growing pains, malignancies, overuse syndromes
Weakness	JDM, myositis secondary to SLE, MCTD, and deep localized scleroderma	Muscular dystrophies, metabolic and other myopathies, hypothyroidism
Chest pain	Juvenile idiopathic arthritis, SLE (with associated pericarditis or costochondritis)	Costochondritis (isolated), rib fracture, viral pericarditis, panic attack, hyperventilation
Back pain	Enthesitis-related arthritis, juvenile ankylosing spondylitis	Vertebral compression fracture, diskitis, intraspinal tumor, spondylolysis, spondylolisthesis, bone marrow–occupying malignancy, pain syndromes, osteomyelitis, muscle spasm, injury
Fatigue	SLE, JDM, MCTD, vasculitis, JIA	Pain syndromes, chronic infections, chronic fatigue syndrome, depression

HSP, Henoch-Schönlein purpura; JDM, juvenile dermatomyositis; JIA, juvenile idiopathic arthritis; MCTD, mixed connective tissue disease; SLE, systemic lupus erythematosus.

*Also known as IgA vasculitis.

Table 194.2**Signs Suggestive of Rheumatic Disease**

SIGN	RHEUMATIC DISEASES	COMMENTS	NONRHEUMATIC CAUSES
Malar rash	SLE, JDM	SLE classically spares nasolabial folds	Sunburn, parvovirus B19 (fifth disease), Kawasaki disease
Oral ulcers	SLE, Behçet disease	Behçet disease also associated with genital ulcers	HSV infection, PFAPA syndrome
Purpuric rash	Vasculitis (e.g., ANCA-associated vasculitis), HSP*	HSP* typically starts as small lesions on lower extremities and buttocks that coalesce	Meningococcemia, thrombocytopenia, clotting disorders
Gotttron papules	JDM	Look for associated heliotrope rash, periungual telangiectasias	Psoriasis, eczema
Arthritis	Juvenile idiopathic arthritis, SLE, vasculitis, HSP*, MCTD, scleroderma, acute rheumatic fever, reactive arthritis	Chronic joint swelling (>6wk) required for diagnosis of JIA; MCTD associated with diffuse puffiness of hands	Postviral arthritis, reactive arthritis, trauma, infection, Lyme disease, Kawasaki disease, malignancy, overuse syndromes

ANCA, Antineutrophil cytoplasmic antibody; HSP, Henoch-Schönlein purpura; HSV, herpes simplex virus; JIA, juvenile idiopathic arthritis; JDM, juvenile dermatomyositis; MCTD, mixed connective tissue disease; PFAPA, periodic fever, aphthous stomatitis, pharyngitis, and adenitis; SLE, systemic lupus erythematosus.

*Also known as IgA vasculitis.

Conti...

➤ **Arthritis** is defined by the presence of **intraarticular swelling** or **two** or more of the following findings on joint examination:

- ✓ Pain on motion
- ✓ Loss of motion
- ✓ Erythema
- ✓ heat. (hot to touch)

Conti...

- **Nonrheumatic causes of arthritis** include
 - ✓ **Malignancy**
 - ✓ Septic arthritis
 - ✓ **Lyme disease**
 - ✓ **Osteomyelitis**
 - ✓ Viral infections (e.g., rubella, hepatitis B, parvovirus B19, chikungunya)
 - ✓ Postinfectious etiologies such as Epstein-Barr virus (EBV)
 - ✓ ARF, and
 - ✓ Reactive arthritis.

- **ARF** typically involves a **migratory** (lasting hours to days), painful arthritis.

- **Pain on palpation** of long bones is suggestive of **malignancy**.

Juvenile idiopathic arthritis (JIA)

- JIA is the **most common rheumatic disease** in **children**.
- It's one of the more common chronic illnesses of childhood.
- JIA represents a heterogeneous group of disorders sharing the clinical manifestation of **arthritis**.

EPIDEMIOLOGY

- Incidence 0.8-22.6/100,000 /yr
- Prevalence ranges from 7-401/100,000
- These wide ranging reflect
 - Population differences
 - Environmental exposure
 - Immunogenetic susceptibility
 - Variations in diagnostic criteria

Conti...

- **Oligoarthritis** is the most common subtype (40–50%), followed by
- **Polyarthritis** (25–30%) and
- **Systemic JIA** (5–15%)
- **More females** than males are affected in **both oligoarticular** (3:1) and **polyarticular** (5:1) JIA but are equally affected in systemic JIA (sJIA).

Conti...

- The peak age at onset is **2-4 years** for **oligoarticular disease**.
- Age of onset has a **bimodal distribution in polyarthriti**s, peaks at **2-4 yrs** and **10-14 years**.
- **sJIA** occurs throughout childhood, with a **peak at 1-5 years**.

Conti...

Category	Proportion of patients (%)
*Systemic	5–10
Oligoarthritis	40–50
Persistent oligoarthritis	25–35
*Extended oligoarthritis	15–20
Polyarthritis	30–40
*Rheumatoid factor-negative	25–35
*Rheumatoid factor-positive	5
Psoriatic arthritis	5–10
Enthesitis-related arthritis	5–10
Other (undifferentiated, overlaps more than one category)	10

Conti...

➤ **F**: **M**

- **1:1** in sJIA
- **3** : **1** in oligoarticular JIA
- **5** : **1** in polyarticular JIA
- 1:1 in PA
- 1:**7** ERA

Conti...

➤ The peak age at onset

- **Oligoarticular:**
 - 2nd and 3rd yrs , less common > 5 yrs and rare after 10 yrs.
- **Polyarthriti**s: Bimodal
 - 2-4 yr and 10-14 yr
- **sJIA** : throughout childhood with a peak 1 to 5 yr
- ERA: mean age at onset of 9-12 yr
- PA: onset 7-10yrs

ETIOLOGY

- Not completely understood
- Multifactorial
- Immunogenetic susceptibility
- **Both genetic and environmental factors** involved
- Concordance rates in monozygotic twins - 25% to 40%

PATHOGENESIS

- JIA is an autoimmune disease
- Alterations of humoral and cell-mediated immunity promote inflammation.
- T lymphocytes release proinflammatory cytokines
- B-cell activation
- Immune complex formation
- Complement activation

Conti...

- Immunologic abnormalities cause
 - Inflammatory synovitis
 - Vascular endothelial hyperplasia
 - Pannus formation (extra growth in joints)
 - Erosion of articular cartilage and contiguous bone

Table 196.1

Criteria for the Classification of Juvenile Rheumatoid Arthritis

Age at onset: <16 yr

Arthritis (swelling or effusion, or the presence of two or more of the following signs: limitation of range of motion, tenderness or pain on motion, increased heat) in one or more joints

Duration of disease: ≥ 6 wk

Onset type defined by type of articular involvement in the first 6 mo after onset:

Polyarthritis: five or more inflamed joints

Oligoarthritis: four or fewer inflamed joints

Systemic-onset disease: arthritis with rash and a characteristic quotidian fever

Exclusion of other forms of juvenile arthritis

CLINICAL MANIFESTATIONS

- **Arthritis** must be present **≥6 weeks** to make a diagnosis of any JIA subtype.
- **Arthritis** is defined by **intraarticular swelling** or the presence of two or more of the following signs:
 - ✓ limitation in range of motion (ROM)
 - ✓ Tenderness or pain on motion, and warmth.
- Initial symptoms may be subtle or acute and often include **morning stiffness** with a limp or **gelling after inactivity**.
- Easy fatigability and poor sleep quality may be present.

Conti...

- Involved joints are:-
 - ✓ Swollen
 - ✓ Warm to the touch, and
 - ✓ Uncomfortable on movement or palpation with reduced ROM
 - ✓ Usually are **not erythematous**.
- Arthritis in large joints, especially knees, **initially accelerates linear growth and causes the affected limb to be longer**, resulting in a discrepancy in limb lengths.
- Continued inflammation stimulates rapid and premature closure of the growth plate, resulting in **shortened bones**.

Classification

Table 196.2

International League of Associations for Rheumatology Classification of Juvenile Idiopathic Arthritis (JIA)

CATEGORY	DEFINITION	EXCLUSIONS
Systemic JIA	Arthritis in one or more joints with, or preceded by, fever of ≥ 2 wk in duration that is documented to be daily (quotidian*) for at least 3 days and accompanied by one or more of the following: 1. Evanescent (nonfixed) erythematous rash 2. Generalized lymph node enlargement 3. Hepatomegaly or splenomegaly or both 4. Serositis [†]	a. Psoriasis or a history of psoriasis in patient or first-degree relative b. Arthritis in an HLA-B27–positive male beginning after the sixth birthday c. Ankylosing spondylitis, enthesitis-related arthritis, sacroiliitis with IBD, Reiter syndrome, or acute anterior uveitis, or history of one of these disorders in a first-degree relative d. Presence of IgM RF on at least two occasions at least 3 mo apart
Oligoarthritis	Arthritis affecting one to four joints during the first 6 mo of disease; two subcategories are recognized: 1. Persistent oligoarthritis—affecting four or fewer joints throughout the disease course 2. Extended oligoarthritis—affecting five or more joints after the first 6 mo of disease	a, b, c, d (above) <i>plus</i> e. Presence of systemic JIA in the patient
Polyarthritis (RF negative)	Arthritis affecting five or more joints during the first 6 mo of disease; a test for RF is negative	a, b, c, d, e
Polyarthritis (RF positive)	Arthritis affecting five or more joints during the first 6 mo of disease; two or more tests for RF at least 3 mo apart during the first 6 mo of disease are positive	a, b, c, e
Psoriatic arthritis	Arthritis and psoriasis or arthritis and at least two of the following: 1. Dactylitis [‡] 2. Nail pitting [§] and onycholysis 3. Psoriasis in first-degree relative	b, c, d, e
Enthesitis-related arthritis	Arthritis and enthesitis [¶] or arthritis or enthesitis with at least two of the following: 1. Presence of or history of sacroiliac joint tenderness or inflammatory lumbosacral pain, or both** 2. Presence of HLA-B27 antigen 3. Onset of arthritis in a male >6 yr old 4. Acute (symptomatic) anterior uveitis 5. History of ankylosing spondylitis, enthesitis-related arthritis, sacroiliitis with IBD, Reiter syndrome, or acute anterior uveitis in first-degree relative	a, d, e
Undifferentiated arthritis	Arthritis that fulfills criteria in no category or two or more of the above categories	

Table 196.4 Overview of Main Features of Subtypes of Juvenile Idiopathic Arthritis (JIA)							
ILAR SUBTYPE	PEAK AGE AT ONSET (yr)	FEMALE:MALE RATIO	% OF ALL JIA CASES	ARTHRITIS PATTERN	EXTRAARTICULAR FEATURES	LABORATORY INVESTIGATIONS	NOTES ON THERAPY
Systemic arthritis	1-5	1:1	5-15	Polyarticular, often affecting knees, wrists, and ankles; also fingers, neck, and hips	Daily fever; evanescent rash; pericarditis; pleuritis	Anemia; WBC ↑; ESR ↑; CRP ↑; ferritin ↑; platelets ↑ (normal or ↓ in MAS)	Less responsive to standard treatment with MTX and anti-TNF agents; consider IL-1 or IL-6 inhibitors in resistant cases or as first-line therapy
Oligoarthritis	2-4	3:1	40-50 (but ethnic variation)	Knees ++; ankles, fingers +	Uveitis in 30% of cases	ANA positive in 60%; other test results usually normal; may have mildly ↑ ESR/CRP	NSAIDs and intraarticular corticosteroids; MTX occasionally required
Polyarthritis:							
RF negative	2-4 and 10-14	3:1 and 10:1	20-35	Symmetric or asymmetric; small and large joints; cervical spine; temporomandibular joint	Uveitis in 10%	ANA positive in 40%; RF negative; ESR ↑ or ↑↑; CRP ↑ or normal; mild anemia	Standard therapy with MTX and NSAIDs; then, if nonresponsive, anti-TNF agents or other biologics, including abatacept, indicated as first-line therapy
RF positive	9-12	9:1	<10	Aggressive symmetric polyarthritis	Rheumatoid nodules in 10%; low-grade fever	RF positive; ESR ↑; CRP ↑/normal; mild anemia	Long-term remission unlikely; early aggressive therapy is warranted
Psoriatic arthritis	2-4 and 9-11	2:1	5-10	Asymmetric arthritis of small or medium-sized joints	Uveitis in 10%; psoriasis in 50%	ANA positive in 50%; ESR ↑; CRP ↑ or normal; mild anemia	NSAIDs and intraarticular corticosteroids; MTX, anti-TNF agents
Enthesitis-related arthritis	9-12	1:7	5-10	Predominantly lower limb joints affected; sometimes axial skeleton (but less than in adult, ankylosing spondylitis)	Acute anterior uveitis; association with reactive arthritis and inflammatory bowel disease	80% of patients positive for HLA-B27	NSAIDs and intraarticular corticosteroids; consider sulfasalazine as alternative to MTX; anti-TNF agents

1. Oligoarthritis

- ≤ 4 joints within the 1st 6 mo
 - A. Persistent oligoarthritis: has a better chance of achieving remission
 - B. Extended oligoarthritis: Young age and Females
- The **most common form of JIA**
- The disease's mildest form
- It predominantly affects the **large joints of the lower extremities**, such as the **knees and ankles**.
- Isolated involvement of **upper-extremity large joints is less common**.

Conti...

- Systemic manifestations (**other than uveitis**) are absent
- 15% to 20% of oligoarthritic JIA Pts will develop **uveitis**
- Isolated involvement of the **hip is rare**
- **Positive ANA** increased risk of **uveitis**
- ANA is positive in up to 80%
- The presence of a **positive antinuclear antibody (ANA)** test confers increased risk for **asymptomatic anterior uveitis**, requiring **periodic slit-lamp examination**.
- ANA positivity may also be correlated with **younger age at disease onset, females, asymmetric arthritis**, and **fewer involved joints over time**.

2. Polyarthrititis

- ≥5 joints within the first 6 months of being diagnosed with JIA
- Joint involvement is **symmetric**
- Presence of **ANA** is **unrelated to the severity of arthritis**.
- It affects **both small and large joints** of the hands and feet.
- Usually starts many joints at a time
- More likely to last into adulthood



Conti...

A. RF-positive

- ✓ 5 - 10% of JIA
- ✓ Older children, onset **10–13 years**
- ✓ Symmetric
- ✓ Rheumatic nodules in 10%
- ✓ Large and small joints
- ✓ ANA is positive in 75%
- ✓ F:m=8-9:1
- ✓ **More aggressive**

B. RF Negative

- ✓ More common
- ✓ Fewer joints than RF positive
- ✓ F > M.... F:M=3:1
- ✓ Uveitis = 10%
- ✓ Peak 1–3 years and 9–14 years
- ✓ ANA positive = 25%
- ✓ Symmetric pattern
- ✓ **Better prognosis**

3. SYSTEMIC JIA (sJIA)

- Is characterized by **arthritis, fever, rash, and prominent visceral involvement**, including **hepatosplenomegaly, lymphadenopathy & serositis** (pericarditis).
- The characteristic **fever, defined as spiking temperatures to $\geq 39^{\circ}\text{C}$**
 - ✓ Occurs on a daily or twice-daily basis for at least 2 wk
 - ✓ Rapid return to normal or subnormal temperatures.
 - ✓ Often present in the evening
 - ✓ Is frequently accompanied by a characteristic **faint, erythematous, macular rash**.
 - ✓ The classic rash is **nonpruritic** and **migratory with lesions lasting <1 hour**.

Conti...

- **More severe form of JIA.**
- Unlike other categories of JIA, sJIA is an **autoinflammatory disease**.
- **sJIA is a diagnosis of exclusion.**
- Uniquely characterized with **extra-articular symptoms**



Conti...

- **Koebner phenomenon**, a cutaneous hypersensitivity in **which classic lesions** are brought on by **superficial trauma**, is often present.
- Heat can also evoke rash.
- Fever, rash, hepatosplenomegaly, and lymphadenopathy are present in **>70% of affected children.**

LABORATORY

- JIA is a **clinical diagnosis** without any diagnostic laboratory tests.
- The meticulous **clinical exclusion** of other diseases and many mimics is therefore essential.
- **CBC....↑WBC and ↑ PLTs counts .**
- **Acute phase reactant....↑ ESR and CRP**
- **ANA**
 - Rarely detectable in oligoarthritis
- **Antibodies to histones ...6% of oligarticular JIA, 12% in uveitis**

Conti...

➤ **RF**

- ❖ 5-15% Polyarticular JIA

- ❖ Rare in sJIA

- ❖ Almost always negative in oligoarthritis

➤ **Anti–cyclic citrullinated peptide antibody**

- ❖ Rarely been demonstrated in oligoarthritis

Conti...

- **sJIA** : striking elevations in
 - ❖ **Inflammatory markers**
 - ❖ **White blood cell**
 - ❖ **Platelet counts**
- **Hgb (↓)** :7-10 g/dl-anemia of chronic disease
- **ESR (↑)** except in MAS
- ↑ **IG levels**
- Ferritin typically ↑,markedly ↑ in MAS (>10,000 ng/mL)

Conti...

➤ **Radiographic:**

- ❖ Soft tissue swelling

- ❖ Periarticular osteopenia

- ❖ Periosteal new-bone apposition

➤ **MRI** is more sensitive than radiography to detect early changes

DDx

❖ Septic arthritis

❖ Reactive arthritis

❖ ALL

❖ SLE

❖ Trauma or nonaccidental injury

❖ Osteomyelitis

❖ Bone tumours

❖ IBD

❖ HSP

❖ Vasculidites

❖ Rheumatic fever

❖ Hypermobility

MANAGEMENT

- Multidisciplinary involvement
 - ❖ Pediatric rheumatologist
 - ❖ Physiotherapist
 - ❖ Occupational therapist
 - ❖ Mental health worker
 - ❖ Ophthalmologist
 - ❖ Orthopaedic surgeon.

The goals of treatment

- ❖ No cure for JIA

 - ✓ But possible to **control the disease**

- ❖ Achieve disease remission

- ❖ Prevent or halt joint damage

- ❖ Foster normal growth and development

- ❖ Monitoring for potential medication toxicities

PROGNOSIS

- The course of JIA unpredictable
- Some prognostic generalizations can be made on the basis of disease type and course

Predictors of severe and persistent disease

- Young age at onset
- RF seropositivity or rheumatoid nodules
- Presence of anti–cyclic citrullinated peptide antibodies
- Large numbers of affected joints
- Polyarticular distribution of arthritis
- Fever lasting >3 mo
- Increased inflammatory markers for >6 mo.
- Disease involving the hip and hand & wrist
- sJIA

Complications

- **Short stature**
- **Uveitis**
- **Leg-length discrepancy**
-
- Joint contractur
- MAS

- **Growth retardation**
- Low bone mineral density
- Chronic pain syndromes
- Problems with school attendance and socialization

Systemic Lupus Erythematosus

INTRODUCTION

- SLE is a **chronic autoimmune disease**.
- Characterized by:-
 - ✓ **Multisystem inflammation**
 - ✓ Presence of **circulating autoantibodies** directed against self-antigens.
- SLE occurs in both children and adults.

Conti...

- Affecting females of **reproductive age**.
- Every organ may be affected, most frequently involved are the **skin, joints, kidneys, blood-forming cells, blood vessels, and CNS**.
- Compared with adults, **children and adolescents** with SLE have:-
 - ✓ **More severe disease** and
 - ✓ **More widespread organ involvement**.

Etiology

- The pathogenesis of SLE remains largely unknown.
- But several factors likely influence risk and severity of disease, including
 - ✓ Genetics
 - ✓ Hormonal
 - ✓ Environmental exposures.

Conti...

- **A genetic predisposition to SLE** is suggested by the association with specific **genetic abnormalities**, including **congenital deficiencies of C1q, C2, and C4**, as well as **several polymorphisms** (e.g., interferon regulatory factor 5, protein tyrosine phosphatase N22), and familial clustering of SLE or other autoimmune disease.
- In addition, certain **human leukocyte antigen** (HLA) types (including HLA-B8, HLA-DR2, and HLA-DR3) occur with increased frequency in patients with SLE.
- Patients with SLE often have **family members**, especially **mothers & sisters**, with SLE or other autoimmune diseases.

Conti...

- SLE **preferentially affects females**, especially during their **reproductive years**, it is suspected that **hormonal factors** are important in pathogenesis.
- Of individuals with SLE, **90% of the cases are female**.
 - ✓ Making **female sex the strongest risk factor for SLE**.
 - ✓ **Estrogens** are likely to play a role in SLE.

Conti...

- **Environmental exposures** that may trigger the development of SLE remain largely unknown.
 - ✓ Certain **viral infections**, including **Epstein-Barr virus (EBV)**, may play a role in susceptible individuals.
 - ✓ **Ultraviolet light exposure** is known to trigger SLE disease activity.
- Environmental influences also may induce epigenetic modifications to DNA, increasing the risk of SLE and **drug induced lupus**.
- Drugs such as **procainamide** and **hydralazine** can promote lymphocyte hypomethylation, causing a **lupus-like syndrome**.

Epidemiology

- Prevalence of SLE in children and adolescents (1-6/100,000) is lower than that in adults (20-70/100,000).
- Prevalence of SLE is highest among blacks, Asians, Hispanics, Native Americans, and Pacific Islanders for both adult and pediatric populations.
- SLE predominantly affects females, with reported of:-
 - ✓ **2- 5 : 1** ratio **before puberty**.
 - ✓ **9 : 1** ratio during **reproductive years**.
 - ✓ Return to **near prepubertal** ratios in the **postmenopausal period**.

Conti...

- Childhood SLE is **rare before 5 yr of age.**
- Usually diagnosed in adolescence, with a **median age of 11-12 yr.**
- Up to **20%** of all individuals with SLE are diagnosed before age **16 yr.**
- Define **pediatric-onset lupus** as onset of symptoms **before age 16, 18 yrs**

Pathology

- **Histologic features** most suggestive of SLE include findings in the **kidney & skin**.
- Renal manifestations of SLE are classified histologically according to the criteria of the International Society of Nephrology (ISN).
- **Renal biopsies** are helpful to establish the **diagnosis of SLE & to stage disease**.
- **Immune complexes** are typically found with **“full house”** deposition of immunoglobulin and complement.

Conti...

- The characteristic **discoid rash** is characterized on biopsy by **hyperkeratosis**, follicular plugging, and infiltration of mononuclear cells into the **dermal-epidermal junction** (DEJ).
- The histopathology of photosensitive rashes can be nonspecific, but immunofluorescence examination of both affected and nonaffected skin may reveal deposition of immune complexes within the DEJ.
- This finding is called the lupus band test , which is specific for SLE.



FIG. 183.1 Mucocutaneous manifestations of SLE. A, Malar rash; B, vasculitic rash on toes; C, oral mucosal ulcer; D, discoid rash in malar distribution.

Pathogenesis

- A hallmark of SLE is the **generation of autoantibodies directed against self antigens**, particularly **nucleic acids**.
- These **intracellular antigens** are expressed everywhere but are usually inaccessible and cloistered within the cell.
- During **cell necrosis or apoptosis**, the **antigens are released**.
- **SLE skin cells are highly susceptible to damage from ultraviolet (UV) light**, and the resulting cell death leads to **release of cell contents**, including **nucleic antigens**.

Conti...

- Individuals with SLE may have greatly **increased levels of apoptosis or significantly impaired ability to clear cell debris.**
 - ✓ Causing prolonged exposure to **nucleic antigens in the bloodstream** and
 - ✓ Increased opportunity for **recognition by immune cells**, leading to **B-cell stimulation** and **autoantibody production**.
- **Circulating autoantibodies form immune complexes** and deposit in tissues, leading to local complement activation, initiation of a proinflammatory cascade, and, ultimately, tissue damage.

Conti...

- **Antibodies to double-stranded (ds) DNA** can form **immune complexes**, deposit in glomeruli, and **initiate inflammation leading to glomerulonephritis**.
- However, many individuals with SLE have circulating antibodies to dsDNA but do not have nephritis, suggesting that autoantibodies are not the only pathway leading to end-organ damage in SLE.
- Both the innate and the adaptive arms of the immune system have been implicated in the dysregulation of the immune system seen in SLE.
- In **active SLE, B-cell populations have impaired tolerance and increased autoreactivity**, enhancing their ability to produce autoantibodies after exposure to self-antigen.

CLINICAL MANIFESTATIONS

- Any organ system can be involved in SLE, so the potential clinical manifestations are myriad.
- The presentation of SLE in childhood or adolescence differs somewhat from that seen in adults.
- The most common presenting complaints of children with SLE include **fever, fatigue, hematologic abnormalities, arthralgia, and arthritis.**
- Many children and adolescents have an **insidious onset of persistent fever**, weight loss, fatigue, and arthralgia, with general deterioration over weeks to months

Conti...

- Arthritis is usually present in the first year of diagnosis.
- Arthritis may be painful or painless swelling, often with stiffness in the morning, and is usually a symmetric polyarthritis affecting large and small joints.
- Tenosynovitis is often present, but joint erosions or other radiographic changes are rare.

Conti...

- Renal disease in SLE is often asymptomatic, underscoring the need for careful monitoring of blood pressure and urinalyses.
- In adolescents, SLE can present with **nephrotic syndrome** and/or **renal failure**, with the predominant symptoms being edema, fatigue, changes in urine color, and nausea/vomiting.
- Because SLE symptoms and findings may develop serially over several years and not all be present simultaneously, the **diagnosis may require longitudinal follow-up.**
- The **neuropsychiatric complications** of SLE may occur with or without apparently active SLE, posing a particularly difficult diagnostic challenge in adolescents, who are already at high **risk for mood disorders.**

Table 199.2**Potential Clinical Manifestations of Systemic Lupus Erythematosus**

TARGET ORGAN	POTENTIAL CLINICAL MANIFESTATIONS
Constitutional	Fatigue, anorexia, weight loss, fever, lymphadenopathy
Musculoskeletal	Arthritis, myositis, tendonitis, arthralgias, myalgias, avascular necrosis, osteoporosis
Skin	Malar rash, discoid (annular) rash, photosensitive rash, cutaneous vasculitis (petechiae, palpable purpura, digit ulcers, gangrene, urticaria), livedo reticularis, periungual capillary abnormalities, Raynaud phenomenon, alopecia, oral and nasal ulcers, panniculitis, chilblains, alopecia
Renal	Hypertension, proteinuria, hematuria, edema, nephrotic syndrome, renal failure
Cardiovascular	Pericarditis, myocarditis, conduction system abnormalities, Libman-Sacks endocarditis

Neurologic	Seizures, psychosis, cerebritis, stroke, transverse myelitis, depression, cognitive impairment, headaches, migraines, pseudotumor, peripheral neuropathy (mononeuritis multiplex), chorea, optic neuritis, cranial nerve palsies, acute confusional states, dural and cerebral venous sinus thrombosis
Pulmonary	Pleuritis, interstitial lung disease, pulmonary hemorrhage, pulmonary hypertension, pulmonary embolism
Hematologic	Immune-mediated cytopenias (hemolytic anemia, thrombocytopenia or leukopenia), anemia of chronic inflammation, hypercoagulability, thrombocytopenic thrombotic microangiopathy, macrophage activation syndrome
Gastroenterology	Hepatosplenomegaly, pancreatitis, vasculitis affecting bowel, protein-losing enteropathy, peritonitis
Ocular	Retinal vasculitis, scleritis, episcleritis, papilledema, dry eyes, optic neuritis

DIAGNOSIS

➤ The diagnosis of SLE requires :-

- ✓ **Comprehensive clinical and laboratory assessment** revealing characteristic multisystem disease and
- ✓ **Excluding** other etiologies, including infection and malignancy.

Conti...

- Over the past several decades, SLE classification criteria have undergone serial updates.
- Both the **American College of Rheumatology (ACR) 1997 Revised Classification Criteria for SLE** and the **2012 Systemic Lupus International Collaborating Clinics (SLICC) Classification** are validated in childhood onset SLE populations, with the **SLICC Criteria achieving higher sensitivity** and the **1997 ACR Criteria achieving better specificity**.
- In both the 1997 ACR and 2012 SLICC Criteria, a **positive antinuclear antibody (ANA) test result is not required for the diagnosis of SLE**; however, ANA negative lupus is extremely rare.

Conti...

- **The ANA test is very sensitive** for SLE (95-99%) but is not very specific (~50%).
- The ANA may become positive many years before a diagnosis of SLE is established; however, most asymptomatic ANA-positive patients will never develop SLE.
- More recently, the 2019 European League Against Rheumatism/American College of Rheumatology (EULAR/ACR) SLE Classification Criteria provided a different approach by requiring a positive antinuclear antibody (ANA) test result at a titer $\geq 1:80$.
- All though the **2019 EULAR/ACR criteria have excellent sensitivity and specificity** in the adult SLE population, they are still being **validated in pediatric populations**.

Conti...

- **Anti-dsDNA and anti-Smith antibodies** are specific for SLE (~98%) but not as sensitive (40–65%) as the ANA.
- **Hypocomplementemia** was not included in earlier classification criteria, but has been added to the two most recent classification criteria.
- The 2012 SLICC Criteria also added as criteria nonscarring alopecia, additional cutaneous and neurologic manifestations, and a positive direct Coombs test in the absence of hemolytic anemia.

Entry criterion
Antinuclear antibodies (ANA) at a titer of $\geq 1:80$ on HEp-2 cells or an equivalent positive test (ever)

If absent, do not classify as SLE
If present, apply additive criteria

Additive criteria

- Do not count a criterion if there is a more likely explanation than SLE.
- Occurrence of a criterion on at least one occasion is sufficient.
- SLE classification requires at least one clinical criterion and ≥ 10 points.
- Criteria need not occur simultaneously.
- Within each domain, only the highest weighted criterion is counted toward the total score.*

Clinical domains and criteria	Weight	Immunology domains and criteria	Weight
Constitutional Fever	2	Antiphospholipid antibodies Anti-cardiolipin antibodies OR Anti- $\beta 2$ GP1 antibodies OR Lupus anticoagulant	2
Hematologic Leukopenia	3	Complement proteins Low C3 OR low C4	3
Thrombocytopenia	4	Low C3 AND low C4	4
Autoimmune hemolysis	4	SLE-specific antibodies Anti-dsDNA antibody [†] OR Anti-Smith antibody	6
Neuropsychiatric Delirium	2		
Psychosis	3		
Seizure	5		
Mucocutaneous Non-scarring alopecia	2		
Oral ulcers	2		
Subacute cutaneous OR discoid lupus	4		
Acute cutaneous lupus	6		
Serosal Pleural or pericardial effusion	5		
Acute pericarditis	6		
Musculoskeletal Joint involvement	6		
Renal Proteinuria >0.5 g/24hr	4		
Renal biopsy class II or V lupus nephritis	8		
Renal biopsy class III or IV lupus nephritis	10		
Total score:			

Classify as systemic lupus erythematosus with a score of 10 or more if entry criterion fulfilled.

Conti...

- For the **older ACR criteria**, a patient is classified as having SLE if **4 or more of the criteria are present**, either serially or simultaneously during any interval of observation in the absence of another explanation for the findings.
- Similarly, for the **SLICC criteria**, **4 or more of the criteria are required** for classification, but the criteria present must include at **least one clinical and one immunologic criterion**.
- An alternative to the requirement of 4 criteria is positive antinuclear antibodies (ANAs) or anti-double-stranded DNA antibodies (anti-dsDNA) in conjunction with biopsy-proven lupus nephritis (LN).

Conti...

- The 2019 joint EULAR/ACR classification criteria are notably different in that a **positive ANA is required for further consideration of SLE.**
- As seen in the table, the criteria are weighted within domains, providing a clinician a sense of the relative likelihood of SLE, depending upon the clinical or laboratory feature present within a domain.
- **A total score of 10 or greater classifies a patient as having SLE.**
- As an example, a patient with mucocutaneous features might have nonscarring alopecia (score 2), oral ulcers (score 2), subacute cutaneous lupus or discoid lupus (either scores 4), or acute cutaneous lupus (ie, malar rash; score 6). **Only the highest score** is counted within each domain, so a child with a malar rash and an oral ulcer would score 6.

Table 199.6**Autoantibodies Commonly Associated with Systemic Lupus Erythematosus (SLE)**

ANTIBODY	CLINICAL ASSOCIATION
Anti-double-stranded DNA (anti-dsDNA)	Correlates with disease activity, especially nephritis, in some with SLE
Anti-Smith antibody (anti-Sm)	Specific for the diagnosis of SLE
Anti-ribonucleoprotein antibody (anti-RNP)	Increased risk for Raynaud phenomenon, interstitial lung disease, and pulmonary hypertension
Anti-Ro antibody (anti-SSA) Anti-La antibody (anti-SSB)	Associated with sicca syndrome May suggest diagnosis of Sjögren syndrome Increased risk of neonatal lupus in offspring (congenital heart block) May be associated with cutaneous and pulmonary manifestations of SLE May be associated with isolated discoid lupus
Antiphospholipid antibodies (including lupus anticoagulant and anti-cardiolipin or anti-beta-2 glycoprotein antibodies)	Increased risk for venous and arterial thrombotic events
Antihistone antibodies	Present in a majority of patients with drug-induced lupus May be present in SLE

LABORATORY FINDINGS

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- **A positive ANA test is present in 95–99% of SLE patients**
 - ✓ ANA-negative lupus is extremely rare.
 - ✓ ANA has poor specificity (50%) for SLE
 - ✓ up to 20% of healthy individuals also have a positive ANA test
 - ✓ ANA titers do not correlate with disease activity

Conti...

➤ anti-dsDNA levels

- ✓ *Specific (98%)* but not as sensitive (40–65%)
- ✓ **Correlate with disease activity**, particularly in those with significant nephritis

➤ Anti-Smith antibody

- ✓ *Specific (98%)* but not as sensitive (40–65%)
- ✓ Does not correlate with disease activity

Conti...

- Serum levels of total hemolytic **complement (CH₅₀)**, **C3**, and **C4** are **typically decreased** in active disease and often improve with treatment.
- Hypergammaglobulinemia is a common but nonspecific finding.
- **Inflammatory markers**, particularly the **ESR**, are often elevated in active disease.
- C-reactive protein (CRP) correlates less well with disease activity; significantly elevated CRP values often reflect infection, whereas chronic mild elevation of CRP may indicate increased cardiovascular risk.

Conti...

- **Antiphospholipid antibodies**, which increase clotting risk, can be found in up to 66% of children and adolescents with SLE.
- The primary antiphospholipid antibodies are lupus anticoagulant, anticardiolipin, and anti-beta-2 glycoprotein antibodies.
- When an arterial or venous clotting event occurs in the presence of an antiphospholipid antibody, **antiphospholipid antibody syndrome** is diagnosed.
- Antiphospholipid antibody syndrome can occur in the context of SLE (secondary) or independent of SLE (primary).
- Rarely, antiphospholipid antibodies can result in catastrophic antiphospholipid syndrome, a condition in which clots affect three or more organs/tissues simultaneously; this condition has a very high mortality rate.

Complications

➤ **Medication toxicities**

- ✓ cataracts, avascular necrosis, and osteoporosis.

➤ **Infections.**

- ✓ Bacterial infection (most common)
- ✓ CMV and aspergillosis

- **MAS** is a rare but serious complication

➤ **Coagulopathy**

- 1) Hypocoagulable states
- 2) Hypercoagulable states:

➤ **Atherosclerosis**

Management

➤ The goals of therapy for patients with SLE are to:-

- ✓ *Ensure long-term survival*
- ✓ *Achieve the lowest possible disease activity*
- ✓ *Prevent organ damage*
- ✓ *Minimize drug toxicity*
- ✓ *Improve quality of life*
- ✓ *Educate patients and their families about their role in disease management*

REFERENCE

- NELSON TEXT BOOK OF PEDIATRICS 22 EDITION
- UPTO DATE 2025